Question and design

Does model width improve contact accuracy at the same one-epoch exposure? Baseline: exp277, native plus MPNN corpus, 24 layers. Hidden width 2048 -> 4096; MLP width 8192 -> 16384. Keep 32 query / 8 KV heads; head width 64 -> 128. Keep data, optimizer schedule, seeds and global batch fixed.

Architecture verified

Actual Levanter parameter-tree counts: Baseline: 1,471,374,336 parameters. Wider model: 5,861,984,256 parameters (3.984x). 12 configuration, tokenizer and finite-epoch dataset tests passed. Count reproduction: count_parameters.py via the README command.

Current status

64-H100 smoke active in CoreWeave US-EAST-02A. Resized from 128 GPUs after concurrent jobs reduced free capacity. Global batch remains 128 through gradient accumulation. Pinned tokenizer fix verified; distributed initialization succeeded. Checking full-corpus packing, then ten updates and checkpoint output. Production training and contact results are still pending.